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COURSE 1.7 / COMPANION WORKBOOK

Run DHCP on SBD-DHCP01

Workstation leases, secure name registration & an audited lease trail

Design and build the workstation VLAN's addressing service on SBD-DHCP01: a full-subnet scope with exclusions and the three options clients trust, directory authorization, the core's relay verified rather than built, client names registered under one dedicated low-privilege account with name protection, a lease history kept beyond Windows' weekly overwrite, backups, and a proof that a name takeover fails.

MATERIALS EDITION 1.0 / COURSE 1.7

Updated 14 October 2026 · sbd-dhcp

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START HERE / YOUR LEARNING PATH

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Read each lesson alongside the interactive guide and the video. Use the record tables to write down what you actually observe — the workbook is your evidence, not the guide.

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Course review and sign-off

> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

Before you start

SBD-DHCP01 hands the workstation VLAN its addresses, and with them the gateway, the DNS servers, and the domain name every client will trust. This course designs that service, builds it on a member server, registers client names under one dedicated low-privilege account, keeps its lease trail and backups, and proves it with a test client. It runs before the tiered administration plane exists: Tier 0 steps happen at SBD-DC01's console, and SBD-DHCP01 is run at its own console by a server-administration group that course 1.8 adopts as Tier 1. The core's DHCP relay is part of the network's build, so this course verifies it and never changes the router. Evidence goes in C:\Evidence\dhcp — on SBD-DHCP01 for the server's own records, and on SBD-DC01 for the Tier 0 steps — and course 1.8 carries each to its tier's seat. Scope values are sample values from Inside the Network.

Prerequisites

- ☐ Complete courses 1.1–1.6: both domain controllers with DNS (course 1.2) and the domain time hierarchy (course 1.3), course 1.4's DISA media set and baseline method, course 1.5's OU structure with the member-server STIG policies on the Servers OU, and course 1.6's PKI. The network's routing, default-deny segmentation and service-path register are already in place, including the core's DHCP relay for VLAN 80.
- ☐ Provision a Windows Server virtual machine for SBD-DHCP01 (10.0.20.10, VLAN 20) with a second volume for its backups and audit-log copies (D: in this sample), and a disposable Windows client virtual machine on VLAN 80 for testing, named SBD-TEST80 in this sample, with sbd.test as its primary DNS suffix and left out of the domain. Turn off the hypervisor's time synchronization on both, as course 1.3 requires of every virtual machine.
- ☐ Until course 1.8 builds the administration seats, Tier 0 steps (the directory, DNS, and authorizing the server, which takes an Enterprise Admins member) happen at SBD-DC01's console, and SBD-DHCP01 is built and run at its own console with the server-administration accounts lesson 2 creates.

What this course builds

- Design the VLAN 80 scope, its exclusions, lease length, and options, and confirm the core's relay under F04.
- Build and authorize SBD-DHCP01 on a member server run by its own server-administration accounts.
- Register client names under one dedicated account with name protection, never through DnsUpdateProxy.
- Keep the lease history and backups, and prove the service, including a refused name takeover.

Systems in scope

RECORD	YOUR OBSERVATION
SBD-DHCP01 — name, address, OS build	

LESSON 01 / ABOUT 25 MIN

Design the addressing service

WHY THIS MATTERS DHCP decides the first things every workstation believes: its address, its gateway, and which servers answer its names. Design that service before building it — one scope for VLAN 80, the options clients will trust, and the identities that authorize, run, and register for it — and confirm the core already relays VLAN 80's requests.

In this lesson you will

- Decide which segment gets leases and design its scope: range, exclusions, reservations, and lease length.
- Choose the options clients will trust and the identities that authorize, run, and register for the server.
- Confirm the register rows and the core's relay for VLAN 80 without changing the router.

Follow the build

1. Decide where DHCP runs and for whom. Only VLAN 80 gets leases — the 50 PCs, and in this course a temporary test client; every server, every admin seat, and the core keep static addresses, because the register names them by address. SBD-DHCP01 is a member server at 10.0.20.10 on VLAN 20, never a domain controller: a DHCP server on a controller inherits the controller's right to change any record in a secure DNS zone.
2. Design the scope the way Microsoft recommends: the whole subnet, 10.0.80.1–254 with mask 255.255.255.0, then exclusion ranges for what must never be leased — 10.0.80.1–99, which holds the core's gateway at .1, and 10.0.80.200–254 — leaving the pool at 10.0.80.100–199, as Inside the Network records it. One hundred addresses cover the fifty PCs, test clients, and replacement machines whose old leases haven't expired. Plan no reservations: a reservation is keyed to a MAC address that any device can claim, so it is an addressing convenience, never an identity check; if a device ever needs one, record it with an owner and a reason.
3. Set the lease length as the outage budget: with one DHCP server, a lease is how long clients keep working while SBD-DHCP01 is down. Keep the Windows default of 8 days as the sample value — clients try to renew halfway through (day 4) and rebind at seven-eighths (day 7) — and record your own value with that reason. Then choose the options, because every option is something every client trusts: router (option 3) 10.0.80.1, DNS servers (option 6) 10.0.10.10 and 10.0.10.11, both domain controllers and never an outside resolver, and DNS domain name (option 15) sbd.test. Add nothing else: no WINS, no boot server, no proxy auto-discovery.
4. Assign three identities. Authorizing a DHCP server writes to the forest's configuration partition, so an Enterprise Admins member does it — Tier 0 work, done at SBD-DC01's console until course 1.8 builds AdminJumpBox01. The server is built and run by named server-administration accounts in a new group, SBD Tier 1 Admins in this sample, because course 1.8 makes member-server administration Tier 1; and client names are registered by a third, unprivileged account in lesson 3.
5. Confirm the paths. Rule F04 in Inside the Network lets VLAN 80 reach SBD-DHCP01 on UDP 67/68 through the core's DHCP relay, owned by the network owner; the relay is part of the network's build, so verify it from the network's record with the network owner — it must forward to 10.0.20.10 and nowhere else, which lesson 2's first lease then proves — and take any gap to the network owner rather than

changing the router. As a domain member, SBD-DHCP01 also uses F01 (DNS), F02 (Kerberos), F03 (directory), and F09 (time); its registrations ride F01, and A02 already lists it for course 1.8's Tier 1 seat.

Check your result

```
# At SBD-DC01's console (Tier 0): the values and identities the design names
Resolve-DnsName SBD-DC01.sbd.test -Server 10.0.10.11; Resolve-DnsName SBD-DC02.sbd.test -
Server 10.0.10.10
Get-ADOrganizationalUnit -Identity 'OU=Servers,OU=SBD,DC=sbd,DC=test' | Select-Object
Name, DistinguishedName
Get-ADGroupMember 'Enterprise Admins' | Select-Object Name   # who can authorize a DHCP
server
Get-ADGroupMember DnsUpdateProxy   # expected: no members
Get-DnsServerZone -ComputerName SBD-DC01 | Select-Object ZoneName, IsDsIntegrated,
DynamicUpdate, ReplicationScope
```

> **WHAT SUCCESS LOOKS LIKE** The design records VLAN 80 as the only leased segment, the full-subnet scope with its two exclusion ranges and no reservations, the lease length with its reason, the three options, who holds each of the three identities, and the network's confirmation that the core relays VLAN 80 to 10.0.20.10 under F04.

Your record

Evidence to keep: Keep the DHCP design (scope, exclusions, lease length, options, identities), the relay confirmation with where it came from, and the register rows that name SBD-DHCP01.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why does SBD-DHCP01 run on a member server instead of a domain controller?

> **REMEMBER** One leased segment, VLAN 80, with a full-subnet scope, two exclusions, and no reservations. The lease length is the outage budget, and three options are all clients get. Authorization, administration, and registration each have their own identity; the relay is confirmed, not built.

Microsoft · Dynamic Host Configuration Protocol (DHCP)

Microsoft · DHCP scopes

Microsoft · Troubleshooting guide: DHCP authorization failures

Microsoft · DNS dynamic update

NIST · SP 800-53 Rev. 5 security controls

LESSON 02 / ABOUT 30 MIN

Build SBD-DHCP01 and its scope

> **WHY THIS MATTERS** Addressing is the first service fifty machines will ask for. Build the DHCP server on a member server run by its own administration accounts, authorize it in the directory — the check that keeps unauthorized Windows DHCP servers from serving — and create the VLAN 80 scope, then pull a first lease through the core's relay.

In this lesson you will

- Build SBD-DHCP01 in the Servers OU, run by SBD Tier 1 Admins.
- Install and authorize the DHCP role, then create the VLAN 80 scope with the planned exclusions and options.
- Prove a first lease through the core's relay before any name is registered.

Follow the build

1. At SBD-DC01's console, create SBD Tier 1 Admins and a named server-administration account for each person who will run SBD-DHCP01, all in the Admin OU, and let the group create computer objects in the Servers OU. The delegation covers that OU and computer objects only, so the group can add servers but gains no rights over the ones already there, including SBD-SUBCA01 and SBD-PKI-MGMT01, which course 1.8 moves to their own OU.
2. Install Windows Server on SBD-DHCP01 with its static address (10.0.20.10/24, the core's 10.0.20.1 as gateway, both domain controllers as DNS servers) and its hypervisor time synchronization off, then, at its console, join it straight into the Servers OU with Add-Computer -OUPath and a server-administration account, so course 1.5's member-server policies apply from first boot. After the restart, still as the local Administrator (the STIG policy may have renamed it), make SBD Tier 1 Admins the only domain group in its local Administrators group, in place of Domain Admins, as the member-server STIG expects; from then on, work as your server-administration account.
3. Install the DHCP role with its tools, create the DHCP Administrators and DHCP Users groups, and restart the service. An Enterprise Admins member then authorizes the server at SBD-DC01's console, after adding the DHCP management tools there (course 1.8 moves this work to AdminJumpBox01). Restart the DHCP Server service on SBD-DHCP01 so it checks its authorization now, and find event 1044: authorized, and servicing clients.
4. Create the scope inactive, exactly as designed: 10.0.80.1–254, mask 255.255.255.0, the two exclusion ranges, an 8-day lease, and the router, DNS server, and domain name options. Turn dynamic DNS updates off for now (DynamicUpdates Never): a record registered under the server's own computer account stays owned by that account, so lesson 3's dedicated account could never update it. Then activate the scope.
5. Verify the core's relay with the test client: on SBD-TEST80 in VLAN 80, release and renew, and confirm the lease came from 10.0.20.10 with the planned gateway, DNS servers, and suffix. The relay wrote its own VLAN 80 address, 10.0.80.1, into each request's gateway address field; that is how SBD-DHCP01 picked the 10.0.80.0 scope. On SBD-DHCP01, find the lease in the scope — and no DNS record yet, by design.

Check your result

```
# At SBD-DC01's console (Tier 0): the group and its one delegation; then create each
person's named
# server-administration account in the Admin OU and add it to the group (New-ADUser, Add-
ADGroupMember)
New-ADGroup 'SBD Tier 1 Admins' -GroupScope Global -Path 'OU=Admin,OU=SBD,DC=sbd,DC=test'
dsacls 'OU=Servers,OU=SBD,DC=sbd,DC=test' /G 'SBD\SBD Tier 1 Admins:CC;computer'
# On SBD-DHCP01 as its local Administrator: join with a server-administration account,
then, after the restart:
Add-Computer -DomainName sbd.test -OUPath 'OU=Servers,OU=SBD,DC=sbd,DC=test' -Credential
(Get-Credential) -Restart
Add-LocalGroupMember Administrators -Member 'SBD\SBD Tier 1 Admins'; Remove-
LocalGroupMember Administrators -Member 'SBD\Domain Admins'
# As a server-administration account: the role and its local groups
Install-WindowsFeature DHCP -IncludeManagementTools; Add-DhcpServerSecurityGroup; Restart-
Service DHCPServer
# At SBD-DC01's console, as an Enterprise Admins member: authorize
Install-WindowsFeature RSAT-DHCP; Add-DhcpServerInDC -DnsName SBD-DHCP01.sbd.test -
IPAddress 10.0.20.10
# Back on SBD-DHCP01: the authorization check, then the scope, inactive until it is
complete
Restart-Service DHCPServer; Get-DhcpServerInDC
Get-WinEvent -FilterHashtable @{ProviderName='Microsoft-Windows-DHCP-Server'; Id=1044} -
MaxEvents 1
Add-DhcpServerv4Scope -Name 'VLAN 80 workstations' -StartRange 10.0.80.1 -EndRange
10.0.80.254 -SubnetMask 255.255.255.0 -LeaseDuration 8.00:00:00 -State InActive
Add-DhcpServerv4ExclusionRange -ScopeId 10.0.80.0 -StartRange 10.0.80.1 -EndRange
10.0.80.99
Add-DhcpServerv4ExclusionRange -ScopeId 10.0.80.0 -StartRange 10.0.80.200 -EndRange
10.0.80.254
Set-DhcpServerv4OptionValue -ScopeId 10.0.80.0 -Router 10.0.80.1 -DnsServer 10.0.10.10,
10.0.10.11 -DnsDomain sbd.test
Set-DhcpServerv4DnsSetting -DynamicUpdates Never   # lesson 3 turns registration on
Set-DhcpServerv4Scope -ScopeId 10.0.80.0 -State Active
Get-DhcpServerv4Lease -ScopeId 10.0.80.0   # after SBD-TEST80 runs ipconfig /release and
ipconfig /renew
w32tm /query /source   # a domain controller, as course 1.3 designed
```

> **WHAT SUCCESS LOOKS LIKE** Get-DhcpServerInDC lists SBD-DHCP01 at 10.0.20.10 and event 1044 shows it servicing clients; the scope covers 10.0.80.1–254 with exclusions .1–.99 and .200–.254, an 8-day lease, and the three options; DNS updates are off; SBD Tier 1 Admins is the only domain group in its local Administrators; its time comes from a domain controller; and the test client's first lease came from 10.0.20.10 through the relay.

Your record

Evidence to keep: Record the build sheet (OU placement, local Administrators, time source), the authorization output and event 1044, the scope, exclusion, and option configuration, and the test client's ipconfig /all with the matching lease.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	

RECORD	YOUR OBSERVATION
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

What does directory authorization of a DHCP server give you?

> **REMEMBER** SBD-DHCP01 is built in the Servers OU, run by SBD Tier 1 Admins, and authorized. The VLAN 80 scope hands out the planned pool and options; DNS updates wait for lesson 3. A first lease proved the core's relay end to end.

Microsoft · Dynamic Host Configuration Protocol (DHCP)

Microsoft · Deploy DHCP using Windows PowerShell

Microsoft · Troubleshooting guide: DHCP authorization failures

Microsoft · Active Directory domain join permissions

Microsoft · DHCP troubleshooting guidance

Microsoft · DHCP server events

LESSON 03 / ABOUT 30 MIN

Register client names securely

> **WHY THIS MATTERS** Course 1.2's zone accepts only secure updates, so the real question is who writes VLAN 80's records. Give that job to one dedicated, unprivileged account, turn on name protection so no device can take over another's name, and keep the DnsUpdateProxy group empty — it strips the protection off every record its members create.

In this lesson you will

- Create the reverse zone and the dedicated update account, and give SBD-DHCP01 its credential.
- Register A and PTR records for every client, with name protection and cleanup when leases end.
- Explain why DnsUpdateProxy is a trap, and prove the records belong to the update account.

Follow the build

1. At SBD-DC01's console, create the reverse lookup zone for 10.0.80.0/24 — Active Directory-integrated, with the replication scope you chose in course 1.2, and secure updates only — so client PTR records have somewhere to go; course 1.2 created only 10.0.10's.
2. Create the update account there too: a standard user (svc-dhcp-dns in this sample) in the Admin OU and in Domain Users only, with a random password of at least 15 characters, set not to expire, as the STIG allows for an application account whose password its owner changes at least once a year and whenever someone who knew it leaves. It needs no DNS permission of its own: any authenticated account may create a record in a secure zone and becomes the owner of what it creates. Keep DnsUpdateProxy empty: records its members create are left unsecured, so the next device to update the name takes it over.
3. On SBD-DHCP01, store the credential with Set-DhcpServerDnsCredential, then set the server to register both A and PTR records for every client (DynamicUpdates Always), including clients that don't ask (UpdateDnsRRForOlderClients), to remove them when a lease expires or is deleted (DeleteDnsRROnLeaseExpiry), and to protect names (NameProtection). Name protection adds a DHCID record that ties each name to the client that registered it, and an update for that name from a different client fails instead of overwriting it.
4. On SBD-TEST80, renew the lease and watch registration happen: SBD-DHCP01's audit log shows the update requested (30) and done (32), both controllers answer for the name and the address, and a DHCID record sits beside the A record. At SBD-DC01's console, read the record's owner: it must be SBD\svc-dhcp-dns, not SBD-DHCP01's computer account.
5. Record the decision: one update identity for VLAN 80, its owner, where the sealed password lives, and that after any password change you set the credential again on SBD-DHCP01, because registrations fail until you do. With every VLAN 80 record owned by that one account, the server cleans up names when leases end, so VLAN 80 doesn't depend on scavenging, which course 1.2 left as a separate decision.

Check your result

```
# At SBD-DC01's console (Tier 0): the reverse zone (course 1.2's replication scope), the
update account, an empty proxy group
Add-DnsServerPrimaryZone -NetworkId '10.0.80.0/24' -ReplicationScope Domain -DynamicUpdate
Secure
New-ADUser svc-dhcp-dns -Path 'OU=Admin,OU=SBD,DC=sbd,DC=test' -AccountPassword (Read-Host
-AsSecureString) -PasswordNeverExpires $true -Enabled $true
```

```
Get-ADGroupMember DnsUpdateProxy   # expected: no members
# On SBD-DHCP01: the credential, then registration for every client, with name protection
Set-DhcpServerDnsCredential -Credential (Get-Credential 'SBD\svc-dhcp-dns')
Set-DhcpServerv4DnsSetting -DynamicUpdates Always -UpdateDnsRRForOlderClients $true -
DeleteDnsRROnLeaseExpiry $true -NameProtection $true
Get-DhcpServerDnsCredential; Get-DhcpServerv4DnsSetting
# On SBD-TEST80: renew, then ask both controllers for the name, the address, and the DHCID
record
ipconfig /renew
Resolve-DnsName SBD-TEST80.sbd.test -Server 10.0.10.11
Resolve-DnsName (Resolve-DnsName SBD-TEST80.sbd.test -Type A -Server 10.0.10.10).IPAddress
-Type PTR -Server 10.0.10.10
Resolve-DnsName SBD-TEST80.sbd.test -Type DHCID -Server 10.0.10.10
# At SBD-DC01's console: the record's owner (sbd.test lives in DomainDnsZones in this
sample)
Import-Module ActiveDirectory
(Get-Acl 'AD:DC=SBD-
TEST80,DC=sbd.test,CN=MicrosoftDNS,DC=DomainDnsZones,DC=sbd,DC=test').Owner   # SBD\svc-
dhcp-dns
```

> **WHAT SUCCESS LOOKS LIKE** The 10.0.80 reverse zone exists with secure updates; svc-dhcp-dns is in Domain Users only and DnsUpdateProxy is empty; Get-DhcpServerDnsCredential names svc-dhcp-dns; the server registers A and PTR records for every client with name protection and cleanup on; and the test client's A, PTR, and DHCID records answer from both controllers, owned by SBD\svc-dhcp-dns.

Your record

Evidence to keep: Keep the zone and account settings, the empty DnsUpdateProxy result, the server's DNS settings and the credential's account name (never the password), the audit-log lines for the update, both controllers' answers, and the record's owner.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why use a dedicated credential for dynamic DNS updates?

> **REMEMBER** VLAN 80's names are written by one dedicated, unprivileged account, not a computer account. Name protection ties each name to its client, and DnsUpdateProxy stays empty. The test client's records answer from both controllers with the right owner.

Microsoft · Deploy DHCP using Windows PowerShell

Microsoft · DNS dynamic update

Microsoft · Set-DhcpServerv4DnsSetting

Microsoft · Set-DhcpServerDnsCredential

Microsoft · Add-DnsServerPrimaryZone

LESSON 04 / ABOUT 30 MIN

Audit leases and back up the server

> **WHY THIS MATTERS** A router's deny log names addresses, not machines; the DHCP audit log is how you turn an address at a moment in time into a workstation. Keep that log longer than the week Windows keeps it, review who holds leases, know what an unauthorized DHCP server can and can't do here, and back the server up so it can be rebuilt.

In this lesson you will

- Read the DHCP audit log and keep each day's file beyond its weekly reuse.
- Review leases and the authorized-server list, and explain what an unauthorized DHCP server does in a domain.
- Back up the DHCP database and configuration, and record what a restore needs.

Follow the build

1. Read the audit log. Windows writes one file per weekday (DhcpSrvLog-Mon.log, and so on) in %windir%\System32\dhcp, each line carrying an event code, the time, the address, the host name, and the MAC address: 10 for a new lease, 11 a renewal, 12 a release, 30 and 32 a DNS update requested and done, 31 a failed one. Each file is reused a week later, and logging pauses when free disk space drops below the configured minimum, so check Get-DhcpServerAuditLog and keep that volume's free space well above it.
2. Keep every day: a daily task, running as SYSTEM shortly after midnight, copies the previous day's file to D:\DHCP\AuditArchive under its date and appends its SHA-256 hash to a catalog. Keep the copies as long as your records requirement says — a year in this sample, like the audit repository course 1.11 builds — because lease history is what ties an address in any log to a machine.
3. Review the leases each week: scope statistics show how full the pool is, and the lease list should hold only machines you know — today, just the test client. An unknown host name or MAC address is a finding to trace, not something to delete. The server's own event logs record every scope, option, exclusion, and reservation change with the account that made it, and warn when the scope is nearly full.
4. Know what an unauthorized DHCP server can do here. A Windows DHCP server in the domain that isn't authorized refuses to lease and logs event 1046, and a workgroup Windows server stops when it meets an authorized one on its network; a server of any other kind ignores authorization entirely and can hand clients its own gateway and DNS servers, a ready-made interception point. So check that Get-DhcpServerInDC lists SBD-DHCP01 alone and that clients' DHCP server is 10.0.20.10, and record that network controls, such as 802.1X admission in course 1.13, carry the rest.
5. Back up what a rebuild needs. Windows backs the database up every 60 minutes, but to %windir%\System32\dhcp\backup on the same disk, so take a backup now with Backup-DhcpServer, export the configuration and leases with Export-DhcpServer, copy both to D:\Backups\DHCP, and hash them. The DNS update credential isn't in the backup, so the restore steps say to set it again from the sealed record; course 1.19 carries member-server backups like these to the offline cabinet.

Check your result

```
Get-DhcpServerAuditLog   # Enable: True; Path: C:\Windows\system32\dhcp
$dir = Join-Path $env:windir 'System32\dhcp'
```

```
Get-Content (Join-Path $dir ('DhcpSrvLog-{0}.log' -f (Get-Date).ToString('ddd',
[cultureinfo]'en-US'))) -Tail 5
# The daily task's work (SYSTEM, 00:15): yesterday's file kept under its date, with its
hash
New-Item -ItemType Directory D:\DHCP\AuditArchive -Force | Out-Null
$y = (Get-Date).AddDays(-1); $copy = 'D:\DHCP\AuditArchive\DhcpSrvLog-{0:yyyy-MM-dd}.log'
-f $y
Copy-Item (Join-Path $dir ('DhcpSrvLog-{0}.log' -f $y.ToString('ddd', [cultureinfo]'en-
US'))) $copy
Get-FileHash $copy -Algorithm SHA256 | Export-Csv D:\DHCP\AuditArchive\catalog.csv -Append
-NoTypeInformation
# Weekly review: pool use, who holds leases, who may serve
Get-DhcpServerv4ScopeStatistics -ScopeId 10.0.80.0 | Select-Object ScopeId, InUse, Free,
PercentageInUse
Get-DhcpServerv4Lease -ScopeId 10.0.80.0 | Select-Object IPAddress, HostName, ClientId,
AddressState, LeaseExpiryTime
Get-DhcpServerInDC   # SBD-DHCP01 alone
# Backups: the database and a readable export, hashed on the second volume
$b = 'D:\Backups\DHCP\{0:yyyy-MM-dd}' -f (Get-Date); New-Item -ItemType Directory $b -
Force | Out-Null
Backup-DhcpServer -Path (Join-Path $dir 'backup'); Copy-Item (Join-Path $dir 'backup')
(Join-Path $b 'database') -Recurse
Export-DhcpServer -File (Join-Path $b 'SBD-DHCP01.xml') -Leases
Get-ChildItem $b -Recurse -File | Get-FileHash -Algorithm SHA256 | Export-Csv ('{0}-
hashes.csv' -f $b) -NoTypeInformation
```

> **WHAT SUCCESS LOOKS LIKE** Audit logging is on with free space to spare; yesterday's file sits in the archive under its date, with its hash in the catalog; the lease list holds only known machines; Get-DhcpServerInDC lists SBD-DHCP01 alone; and a dated database backup and XML export sit in D:\Backups\DHCP with their hashes.

Your record

Evidence to keep: Keep the audit log settings, the archive catalog, the dated lease review, the authorized-server list, and the backup set with its hash list and restore note.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why copy the DHCP audit log every day?

> **REMEMBER** Each day's lease history is archived with a hash before Windows reuses the file. Leases and the authorized-server list are reviewed; authorization stops only Windows servers. The database and a readable export are backed up, with the credential step written beside them.

Microsoft · DNS dynamic update

Microsoft · DHCP server events

Microsoft · Set-DhcpServerAuditLog

Microsoft · Backup-DhcpServer

Microsoft · Export-DhcpServer

Microsoft · Get-FileHash

LESSON 05 / ABOUT 30 MIN

Prove the service and hand off

> **WHY THIS MATTERS** Prove DHCP the way clients will meet it: the test client gets the right lease and its name registers securely, a device that tries to take that name is refused, and the evidence lands in the logs. Then take SBD-DHCP01's first SCAP baseline and hand the server to course 1.8, which moves its administration to the Tier 1 seat.

In this lesson you will

- Prove the lease, the options, and secure registration from the test client.
- Prove name protection with a deliberate takeover attempt.
- Take SBD-DHCP01's first SCAP baseline, seal the record, and write the handoff to course 1.8.

Follow the build

1. On SBD-TEST80, release and renew, and check everything the client now believes: an address from 10.0.80.100–199, DHCP server 10.0.20.10, gateway 10.0.80.1, DNS servers 10.0.10.10 and 10.0.10.11, suffix sbd.test, and an 8-day lease. On SBD-DHCP01, find the same lease and its audit-log lines, and confirm both controllers answer for the name and the address.
2. Try to take the name: shut the test client down, give it a new MAC address in the hypervisor — to the server, a new client with a new DHCID — and start it again. It gets a lease, but its registration is refused: event 1340 on SBD-DHCP01, a failed update (31) in the audit log, and the name still resolving to the original address. Restore the original MAC address, renew, delete the stray lease, and confirm the name still points at the test client.
3. Take SBD-DHCP01's first SCAP baseline the course 1.4 way: copy SCC and the Windows Server STIG benchmark from the course 1.4 media set into C:\Staging\DISA, check them against the connected-side-hashes.csv that travels with the media, install the same SCC version, and scan locally with the same profile. Store the session unmodified in C:\Evidence\dhcp\baseline, classify each open finding as fix-by-GPO, deviation candidate, or POA&M candidate, and leave the manual checks for a STIG Viewer checklist on the Tier 1 seat in course 1.8.
4. Seal both halves of the record, each into its own manifest: C:\Evidence\dhcp on SBD-DHCP01 (design, build sheet, scope export, registration and takeover results, audit catalog, backup hashes, baseline) and C:\Evidence\dhcp on SBD-DC01 (authorization, reverse zone, update account, record owner). Course 1.8 carries each to its tier's seat.
5. Write the handoff to course 1.8: SBD-DHCP01 serves VLAN 80 and registers names as svc-dhcp-dns; SBD Tier 1 Admins runs it, still at its console; the test client stays for course 1.8's proof, where it joins the domain as the Tier 2 test machine; and course 1.13 adds the Network Policy Server role for 802.1X to this server.

Check your result

```
# On SBD-TEST80: release, renew, and read what the client believes
ipconfig /release; ipconfig /renew; ipconfig /all
# On SBD-DHCP01: the lease, then, after the MAC change, the refused takeover
Get-DhcpServerv4Lease -ScopeId 10.0.80.0 | Select-Object IPAddress, HostName, ClientId,
LeaseExpiryTime
Get-WinEvent -FilterHashtable @{ProviderName='Microsoft-Windows-DHCP-Server'; Id=1340} -
MaxEvents 1 | Format-List TimeCreated, Message
```

```
Resolve-DnsName SBD-TEST80.sbd.test -Server 10.0.10.10   # still the original address
Remove-DhcpServerv4Lease -IPAddress '<stray lease address>'   # after the original MAC is
back
# Before installing SCC: no output means every copied file is on the connected-side list
(Get-ChildItem C:\Staging\DISA\* -File -Exclude *.csv | Get-FileHash -Algorithm
SHA256).Hash | Where-Object { $_ -notin (Import-Csv C:\Staging\DISA\connected-side-
hashes.csv).Hash }
# After the SCC scan: keep the session unmodified (adapt the path to SCC's results
directory), then seal the folder
$session = Get-ChildItem (Join-Path $env:USERPROFILE 'SCC\Sessions') -Directory | Sort-
Object Name | Select-Object -Last 1
New-Item -ItemType Directory C:\Evidence\dhcp\baseline -Force | Out-Null
Copy-Item $session.FullName -Destination C:\Evidence\dhcp\baseline\SBD-DHCP01 -Recurse
Get-ChildItem C:\Evidence\dhcp -Recurse -File | Get-FileHash -Algorithm SHA256 | Export-
Csv C:\Evidence\dhcp-manifest.csv -NoTypeInformation
Get-FileHash -Algorithm SHA256 C:\Evidence\dhcp-manifest.csv
# Run the same two lines at SBD-DC01's console for the Tier 0 half, its own C:\Evidence\
dhcp
```

> **WHAT SUCCESS LOOKS LIKE** The test client holds a lease from the pool with every planned option; its A, PTR, and DHCID records answer from both controllers and belong to SBD\svc-dhcp-dns; the takeover attempt was refused with event 1340 and left the record unchanged; SBD-DHCP01 has a first baseline with every open item classified; and both manifests and the handoff are filed.

Your record

Evidence to keep: Keep the client's ipconfig /all, the lease and audit-log lines, the takeover attempt with event 1340, the SCC session with its hashes and classified findings, both manifests with their own hashes, and the handoff note.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why change the test client's MAC address during the proof?

> **REMEMBER** The test client's lease, options, and secure records are proven from both sides. A name takeover was refused and logged. SBD-DHCP01 is baselined, its record sealed, and course 1.8 takes over its administration.

Microsoft · Set-DhcpServerv4DnsSetting

Microsoft · DHCP server events

DISA · Security Content Automation Protocol (SCC and SCAP benchmarks)

Microsoft · Get-FileHash

NIST · SP 800-53 Rev. 5 security controls

Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

Lesson completion

LESSON	EVIDENCE FILED	DATE
01 Design the addressing service	☐ yes	
02 Build SBD-DHCP01 and its scope	☐ yes	
03 Register client names securely	☐ yes	
04 Audit leases and back up the server	☐ yes	
05 Prove the service and hand off	☐ yes	

Open items carried forward

Next in the learning path

That completes Run DHCP on SBD-DHCP01. Course 1.8 builds the tiered administration seats and moves SBD-DHCP01's administration to SBD-ADMIN-SRV01.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.

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